Supplemental Materials

Algorithm S1. Scenario-constrained stratified sampling

The procedure referred to in Section 3.4 of the main text is stated below. Sampling selects which workers enter the study set; it is applied to the label set and never used to create structure. No node is duplicated or synthesized, and no relationship among the retained workers is synthesized or rewired, so every tie over which messages pass is one observed in the employment register. The graph analyzed is the subgraph induced on the 10826 sampled workers; ties between retained and non-retained workers are therefore absent by construction, and the topology-derived covariates are computed on this induced subgraph.

Input: graph $G = (V, E, \psi)$; labels y; context map C; target ratio $\rho = 3$; stratum shares (0.65, 0.20, 0.15)

Output: labeled index set S used for loss computation

$P \leftarrow \{v \in V : y_v = 1\}$; $n \leftarrow |P|$; $m \leftarrow \rho \cdot n$

$\Omega \leftarrow \{C_v : v \in P\}$ //context cells containing at least one positive

$N_1 \leftarrow$ sample $(\lfloor 0.65m \rfloor$, from $\{v : y_v = 0, C_v \in \Omega\})$ //scenario–constrained hard negatives

for each $v \in P$ do $q(v) \leftarrow$ nearest $y_v = 0$ neighbours of x_v with $C_v \notin \Omega$, within caliper δ

$N_2 \leftarrow$ sample $(\lfloor 0.20m \rfloor$, from $\cup_{v \in P} q(v))$ // feature–matched negatives

$N_3 \leftarrow$ sample $(m - |N_1| - |N_2|$, from $\{v : y_v = 0\} \setminus (N_1 \cup N_2))$ //random negatives

$S \leftarrow P \cup N_1 \cup N_2 \cup N_3$; the graph is left unmodified

return S

Stratum N_1 matches the context vector C exactly, so the residual variation between positives and these negatives is relational rather than environmental. N_2 controls attribute confounding separately from context, using 1:1 nearest-neighbour matching without replacement on Mahalanobis distance over standardized continuous attributes, with exact matching on trade and skill grade and a caliper of 0.2 standard deviations. N3 introduces an unconditioned component drawn from the remaining eligible negative pool. The combined study set does not reproduce the marginal distribution of the full workforce. The natural worker-level prevalence and deployment calibration are not estimated in the present study.

Text S1. The criticality index: rationale and weighting sensitivity

Equation (1) defines the composite criticality index as the unweighted mean of the percentile ranks of a worker on betweenness centrality, PageRank and closeness centrality,

$$\mathrm{Crit}(v) = \tfrac{1}{3}\left[r_{\mathrm{B}}(v) + r_{\mathrm{PR}}(v) + r_{\mathrm{C}}(v)\right], \quad \mathrm{Crit}(v) \in [0, 1]$$

.

Percentile ranks are averaged rather than raw values for two reasons. The three measures are on incomparable scales and all three are heavy-tailed on this graph, so an average of raw values would be dominated by betweenness. And rank averaging is invariant to any monotone rescaling of the underlying measures, so the composite does not depend on how the three are normalized.

Equal weights are used because Section 2.2 gives no basis for preferring one of the three mechanisms over the others. To confirm that the inverted-U result of Section 5.4 is not an artifact of that choice, the analysis was repeated under alternative weightings. Table S5 reports, for each, the decile of the criticality distribution at which the observed participation rate peaks, the rate in that decile, the rates in the top and bottom deciles, and the rank correlation with the equal-weight index. The peak lies in the interior under every weighting; its location moves between roughly the 20th and 60th percentiles, which is why Section 5.4 describes the peak as a region rather than a cut point.

Text S2. Prospective evaluation on the deployment graph: design details

Prediction origins t_0 are placed at the start of every quarter from April 2021 to July 2024 (14 origins). At each origin the node set is every worker with at least one engagement overlapping the preceding 365 days, irrespective of label: between 3.6 and 4.1 $\times 10^5$ workers and 0.9 and 1.1 $\times 10^7$ edges per origin. Ties use only co-presence before t_0 : a workgroup tie requires the same project, subcontractor and team leader with presence overlapping by at least seven days, and a hometown tie the same township and the same project under the same overlap condition, presence on a project being taken as the interval from entry to entry plus recorded attendance days. Attributes and work history are computed from engagements that began before t_0 , with attendance truncated at t_0 , and project-, subcontractor- and workgroup-level incident counts include only incidents recorded before t_0 . The label is a worker's first appearance in the incident register within the 91 days after t_0 ; workers already in the register are kept in the graph but not scored. Betweenness is estimated from 256 pivot sources and closeness is replaced by harmonic closeness estimated from the same pivots; all covariates are rank-normalized within each origin.

Models are trained on the nine origins from April 2021 to April 2023, early-stopped on the July and October 2023 origins, and tested once on the three 2024 origins, so no outcome from the test period reaches training. The test population is every eligible worker at the test origins (1140590 worker-origins, 359 participants, 2992 projects), scored without resampling; at a base rate of about 3.1 per 10,000, AUPRC is reported both directly and relative to the base rate. Within-project AUROC is computed separately in each (origin, project) cell that contains both participants and non-participants (61 cells) and averaged over cells. The project-only baseline scores workers solely by the prior incident history of their current project, subcontractor and workgroup. Because each origin graph is too large for a full-graph forward pass, the graph models are trained in mini-batches with uniform neighbour sampling (fan-outs 15 and 10). Training

negatives are drawn by the 65/20/15 design of Section 3.4 within each training origin, except for the alternatives compared in Table S11. Differences are assessed with the project-level paired bootstrap of Section 5.1 of the main text.

Table S1 Feature dictionary, preprocessing rules and model configuration

Item	Setting
Node features — worker attributes	
Age	Continuous, standardized; no missing values after cleaning; trade-level median retained as the rule
Sex	Binary indicator
Trade	One-hot over 149 categories
Skill grade	Ordinal, treated as continuous
Tenure with group (months)	Continuous, log1p-transformed then standardized
Number of project spells	Continuous, standardized
Contract signed	Binary indicator (38.2% signed)
Cumulative attendance (days)	Continuous, winsorized at the 99th percentile
Hometown region	Embedded, dimension 16; 538 counties, rare categories pooled below 20 workers
Node features — topology-derived covariates	
Betweenness, PageRank, closeness	Exact, computed on the 10826-node graph ; rank-normalized to [0, 1]; on the deployment graphs of Section 5.5, betweenness by pivot sampling and harmonic closeness, both from 256 pivots
Preprocessing and missing data	
Missing-value rule	Continuous: trade-level median with a missingness indicator. Categorical: explicit "unknown" level
Records excluded	5770 duplicate registrations; workers with no resolvable township code (2.2%)
Temporal window	2021-01-01 to 2024-10-31; ties require overlapping engagement within the window
Graph	
Node types / edge types	1 / 2 (workgroup, hometown)
Nodes / edges	10826 / 332448
Edge weighting	Workgroup ties weight 1.0, hometown ties 0.5; minimum overlap 30 days
Architecture and optimization	
Propagation layers L / attention heads H	2/4
Embedding dimension / hidden width	64/128
Dropout / residual connections	0.3/yes
Optimizer / learning rate / weight decay	Adam/5e-4/1e-5
Batch strategy	Full-graph forward pass; loss over the sampled label set
Epochs / early stopping	120 to convergence; patience 20 on validation AUPRC
Cost coefficient λ	3.0, matching the sampled negative-to-positive ratio
Random seeds	5-fold cross-validation over 5 seeds (0–4); reported figures are means, dispersion in Table S3
Decision layer	
Threshold selection	Chosen on the validation split as the highest score giving recall $\geq$ 75% (score 0.4255); not cost-optimized (see Table S8)

Note: the sampling ratio and stratum shares were fixed a priori and are not tuned hyperparameters; Table 2 of the main text reports their sensitivity.

Table S2 Structural statistics of the labor network by relation layer

Statistic	Hometown	Workgroup	All layers
Nodes	10826	10826	10826
Edges	149707	182741	332448
Mean degree	27.66	33.76	61.42
Density	0.002555	0.003119	0.005674
Clustering coefficient	0.8519	0.6425	0.7651
Mean shortest-path length	—	—	4.00
Degree assortativity	1.0000	0.9959	0.8174
Edge-homophily ratio	0.6682	0.6884	0.6793
Connected components	—	—	388 (largest 10,244)

Table S3 Dispersion across random seeds

Model	AUPRC (mean ± SD)	Recall % (mean ± SD)	F2 (mean ± SD)	95% CI, AUPRC
SNA-HGT	0.5341 ± 0.0086	76.54 ± 1.42	0.6325 ± 0.0091	[0.5262, 0.5427]
HGT	0.4950 ± 0.0104	67.67 ± 1.88	0.5891 ± 0.0118	[0.4856, 0.5049]
SNA-GraphSAGE	0.5049 ± 0.0097	67.91 ± 1.75	0.6124 ± 0.0106	[0.4961, 0.5140]
GraphSAGE	0.4815 ± 0.0112	50.00 ± 2.31	0.4917 ± 0.0139	[0.4713, 0.4919]
SNA-GAT	0.4686 ± 0.0118	74.13 ± 2.04	0.6067 ± 0.0127	[0.4578, 0.4793]
GAT	0.4024 ± 0.0131	60.20 ± 2.46	0.5356 ± 0.0148	[0.3906, 0.4145]
SNA-XGBoost	0.3805 ± 0.0090	56.67 ± 1.63	0.5139 ± 0.0101	[0.3724, 0.3888]
XGBoost	0.3356 ± 0.0098	46.22 ± 1.79	0.4094 ± 0.0112	[0.3268, 0.3446]

Note: seed-to-seed dispersion describes algorithmic variance and is not used for inference; the intervals are percentiles over seeds and are reported for completeness only. Differences between models are assessed by the project-level paired bootstrap reported in Table 3 of the main text.

Table S4 Comparison of negative-sampling and imbalance-handling designs

Design	Balances classes	Matches context	Preserves topology	Preserves marginals
Random undersampling	Yes	No	Yes	No
Random oversampling	Yes	No	Yes	No
SMOTE (attribute-space interpolation)	Yes	No	Attributes only	No
GraphSMOTE (synthesizes nodes and edges)	Yes	No	No	No
Neighbor selection (CARE-GNN)	No	No	Alters aggregation	Yes
Scenario-constrained stratified (this study)	Yes	Yes	Yes	No

Table S5 Sensitivity of the inverted-U result to the criticality weighting

Weighting (betweenness, PageRank, closeness)	Peak decile: observed rate (%)	Top / bottom decile rate (%); ρ with equal
Equal (⅓, ⅓, ⅓)	20th–30th: 35.9	18.8 / 14.8; 1.00
Betweenness-dominant (0.6, 0.2, 0.2)	30th–40th: 33.3	19.0 / 14.8; 0.98
PageRank-dominant (0.2, 0.6, 0.2)	30th–40th: 33.0	19.1 / 13.5; 0.98
Closeness-dominant (0.2, 0.2, 0.6)	20th–30th: 34.3	17.5 / 18.6; 0.98
PageRank only	50th–60th: 33.2	22.6 / 13.9; 0.90
Closeness only	20th–30th: 31.2	17.5 / 23.7; 0.88

Note: each row re-ranks workers by a differently weighted composite of the three percentile ranks, then reports the observed participation rate by decile of that score on the study set of 10826 workers (sampled positive-label fraction 24.8%). The peak is interior in every variant. ρ is the Spearman correlation of the variant score with the equal-weighting score of Eq. (1). A betweenness-only variant is not reported: betweenness is zero for a large share of nodes, so its lower deciles are determined by arbitrary tie-breaking.

Table S6 Composition of the employment and collective-incident registers and of the study set

Item	Value
Observation window	January 2021 – October 2024
Employment records (worker–project–period)	2037377
Unique workers (after de-duplication)	1278369
Duplicate registrations removed	5770
Distinct projects (register)	5800
Distinct projects (study set)	888
Distinct labor subcontractors (study set)	1094
Distinct counties / townships of origin (study set)	538 / 1778
Distinct trades (study set)	149
Collective incidents recorded (2021–2024)	251
Workers coded positive (named in ≥1 incident)	2683
Projects on which positive workers were employed (study set)	445
Study set (positives : negatives)	10826 (2683 : 8,143 = 1 : 3.04)
Age, mean (SD) (study set)	40.7 (8.5)
Male share (%) (study set)	89.8
Median attendance days (mean) (study set)	57 (100.2)

Table S7 Design-level comparison of SNA-HGT with representative graph-learning methods

Method	Relation-typed parameters	Global structural information	Imbalance handling	Interpretable structural covariates
GCN	No	L-hop only	Loss weighting	No
GAT	No	L-hop only	Loss weighting	No
GraphSAGE	No	L-hop, sampled	Loss weighting	No
GraphSAGE + GraphSMOTE	No	L-hop, sampled	Synthetic minority nodes/edges	No
HAN (metapath-based)	Analyst-specified metapaths	L-hop along metapaths	Loss weighting	No
HGT	Learned per relation	L-hop only	Loss weighting	No
XGBoost	Not applicable	None	Scale-pos-weight	Only if supplied manually
SNA-HGT (this study)	Learned per relation	Global covariates on induced study-set	Context-matched sampling +	Yes

Method	Relation-typed parameters	Global structural information	Imbalance handling	Interpretable structural covariates
		graph	cost-asymmetric loss	

Table S8 Operating-point sensitivity, expected cost, cumulative gain and lift

Threshold	n flagged	P (%)	R (%)	Gain (%)	Lift	E[C] 5:1	E[C] 10:1	E[C] 20:1
Top 1%	108	87.0	3.5	3.5	3.51	12,959	25,904	51,794
Top 5%	541	71.2	14.3	14.3	2.87	11,646	23,136	46,116
Top 10%	1,083	63.3	25.6	25.6	2.55	10,387	20,377	40,357
Top 20%	2,165	52.8	42.6	42.6	2.13	8,716	16,411	31,801
Top 30%	3,247	46.6	56.4	56.4	1.88	7,590	13,445	25,155
Top 50%	5,413	37.8	76.2	76.2	1.52	6,558	9,748	16,128
Selected operating point	5,437	37.8	76.6	76.6	1.52	6,528	9,673	15,963
Top 70%	7,578	31.7	89.5	89.5	1.28	6,587	7,997	10,817
Top 90%	9,743	26.9	97.6	97.6	1.08	7,444	7,764	8,404
Flag all (100%)	10,826	24.8	100.0	100.0	1.00	8,143	8,143	8,143
Fixed 0.5 threshold	2,361	50.9	44.8	44.8	2.05	8,570	15,980	30,800

Note: All quantities in this table refer to the sampled study set and should not be interpreted as deployment-population costs, precision, or prevalence. E[C] = r·FN + FP with C_FP = 1. On the study set, flagging every worker is cheaper than the selected point for any r above about 7.6, so the selected point is not cost-optimal; it is set for recall under the verification budget (Section 5.4).

Table S9 Observed participation by criticality decile on the deployment graphs

Crit(v) percentile band	Worker-origins	Participants	Rate per 10,000	95% Wilson interval
0–10th	542188	127	2.34	[1.97, 2.79]
10–20th	542194	236	4.35	[3.83, 4.94]
20–30th	542196	307	5.66	[5.06, 6.33]
30–40th	542194	268	4.94	[4.39, 5.57]
40–50th	542193	207	3.82	[3.33, 4.37]
50–60th	542196	193	3.56	[3.09, 4.10]
60–70th	542197	234	4.32	[3.80, 4.91]
70–80th	542193	204	3.76	[3.28, 4.32]
80–90th	542197	187	3.45	[2.99, 3.98]
90–100th	542204	161	2.97	[2.54, 3.46]

Note: Crit(v) (Eq. 1 of the main text) is computed within each origin on the deployment graph, without reference to labels; deciles are formed within origin and pooled over the 14 origins (5421952 worker-origins, 2124 participants). A participant is a worker's first appearance in the incident register within 91 days of the origin.

Table S10 Prospective evaluation on the deployment graph

Model	AUROC	AUPRC ($\times 10^{-4}$)	AUPRC / base rate	Recall@top 10% (%)	Within-project AUROC
Project-only baseline	0.531	3.47	1.10	14.2	0.503
SNA-XGBoost	0.757	10.77	3.42	37.5	0.683
HGT	0.768	13.92	4.42	47.2	0.698
SNA-HGT, single relation	0.795	17.70	5.62	50.6	0.721
SNA-HGT	0.812	20.16	6.40	54.5	0.749

Note: 1140590 worker-origins, 359 participants, 2992 projects; base rate 3.15 per 10,000. Scores are averaged over 5 seeds. Within-project AUROC is averaged over the 61 (origin, project) cells containing both participants and non-participants. The project-only baseline uses only the prior incident history of the worker's current project, subcontractor and workgroup. SNA-XGBoost is the tree-based baseline of Table 1, trained on the same attributes and

covariates.

Table S11 SNA-HGT under alternative negative-sampling designs, evaluated on the same deployment population

Training negatives	AUROC	AUPRC / base rate	Recall@top 10% (%)	Within-project AUROC	ρ with project-only score
All negatives, cost-weighted	0.741	3.95	40.3	0.662	0.44
Random, 1:3	0.823	6.71	56.8	0.681	0.58
Context-matched, 1:3	0.776	5.88	48.3	0.752	0.27
Scenario-constrained 65/20/15 (proposed)	0.812	6.40	54.5	0.749	0.31

Note: test population as in Table S10. ρ is the Spearman correlation between a model's scores and those of the project-only baseline. Within-project AUROC, proposed minus random: 0.068 (95% project-level bootstrap interval [0.021, 0.117]).